# Digital Systems

## Lecture 15 & 16

## Scheduling Processes & Messages

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Following the course of past years by Mike Spivey

# C Pointer Syntax

- Variable containing 1

```
int a = 1;
```

- Variable able to contain address to memory location containing `int`

```
int* b;
```

- Set b to address of a

```
b = &a
```

- Set c to contents of a

```
int c = *b
```

# C Pointer Syntax for Structs

```
typedef struct dst {
  int feet;
  float inch;
} distance;
```

- Struct-typed variable

  `distance a; a.feet = 9001; a.inch = 11;`
- Pointer to `distance`-typed variable, assign address of `a`

  `distance* b; b = &a;`
- Content of address in b

  `(*b).inch == a.inch;  /* Or... */ b->inch == a.inch;`

# Today

**How are processes started?**

**What happens to the Operating System on startup?**

## Starting a Process

The first time a process runs, it is resumed just as if returning from a system call.

So we set up a fake exception frame that invokes the process body when resumed.

- `r0` = integer argument,
- `pc` = process body,
- `lr` = address of exit stub, in case body returns.

# Creating “fake” Stack Frame

```
int start(...) {
    ...
    unsigned *sp = p->sp - FRAME_WORDS;
    memset(sp, 0, 4*FRAME_WORDS);
    // Macros for offsets in stack, ***_SAVE
    sp[PSR_SAVE] = INIT_PSR;  // Thumb bit switched on
    sp[PC_SAVE] = (unsigned) body & ~0x1; // Activate the proc
    sp[LR_SAVE] = (unsigned) exit; // Make it return to exit()
    sp[R0_SAVE] = (unsigned) arg;  // Pass arg in r0
    sp[ERV_SAVE] = MAGIC;
    p->sp = sp;
    ... }
```

## Starting a Process

At this point, every process has a stack frame that:

- will restore the `pc` at the start of the process's function (i.e. it will run the function)
- has values for all state and registers that respects the restoring convention (manual part + hardware part),
- has values in the registers that respect the calling convention of C functions (e.g. integer argument in `r0`).

To start, just initiate context switch by returning from `system_call()`.

- Q: Return value?

## Starting the OS

After `init()` called, all tasks have stack frames that will allow a context switch into them.

```
void __start(void) {
    /* Create idle process */
    ...
    /* Call the application's setup function */
    init();
    /* The main program morphs into the idle process. */
    os_current = idle_proc;
    set_stack(os_current->sp);
    idle_task();
}
```

# Idle Process

```
void idle_task(void) {
  /* Pick a genuine process to run */
  yield();

  /* When there's nothing to do: */
  while (1) pause();
}
```

## Where are we now?

We understand *how* tasks are switched and started.

- OS is an interrupt that performs context switches
  (Q: remind me how)
- During context switch, interrupt stores some state on stack,
  assembly code stores the rest.

# What is left to understand?

**? How does the OS keep track of processes?**

We have already seen that we need to start processes, and keep track of various process-specific quantities (e.g. stack location), in order to do context switches. How does the OS do this?

**? How does the OS decide which process to run next?**

**? How are messages transferred between processes?**

And in particular, how is this done in interrupts?

**? How does the OS keep track of processes?**

We have already seen that we need to start processes, and keep track of various process-specific quantities (e.g. stack location), in order to do context switches. How does the OS do this?

# Process Structure

```
struct _proc {
    int pid;            // Process ID
    char name[16];      // Name for debugging
    unsigned state;     // SENDING, RECEIVING, etc.
    unsigned *sp;       // Saved stack pointer
    int priority;       // Priority: 0 is highest
    proc waiting;       // Processes waiting to send
    int pending;        // Whether interrupt pending
    int msgtype;        // Message to send or receive
    message *msgbuf;    // Pointer to message buffer
    proc next;          // Next process in ready/send queue
};
```

## Possible states

- `DEAD`: process that has exited. It is also the value present in slots of the table that have never been used.
- `ACTIVE`: process is ready to run. The process is either the current process, or the ready queue waiting to become current.
- `SENDING`: process is waiting to send to another; the process will appear in the sending queue of the destination.
- `RECEIVING`: process is waiting to receive a message; the `msgtype` field will specify who may send to it.
- `SENDREC`: call to `sendrec`, combining the functions of send and receive.
- `IDLING`: idle process, which runs when no other process is ready.

# Process Structure

```
typedef struct _proc *proc;
...
static proc os_ptable[NPROCS];
```

In `start(...)`:

```
   proc p = create_proc(name, roundup(stksize, 8));
```

In `create_proc(...)` we

- Initialise all the fields to sensible defaults / based on arguments.
- Find a memory location to initialise the stack (`sbrk(...)`)
- Find a memory location for proc variable (`new_proc(...)`)

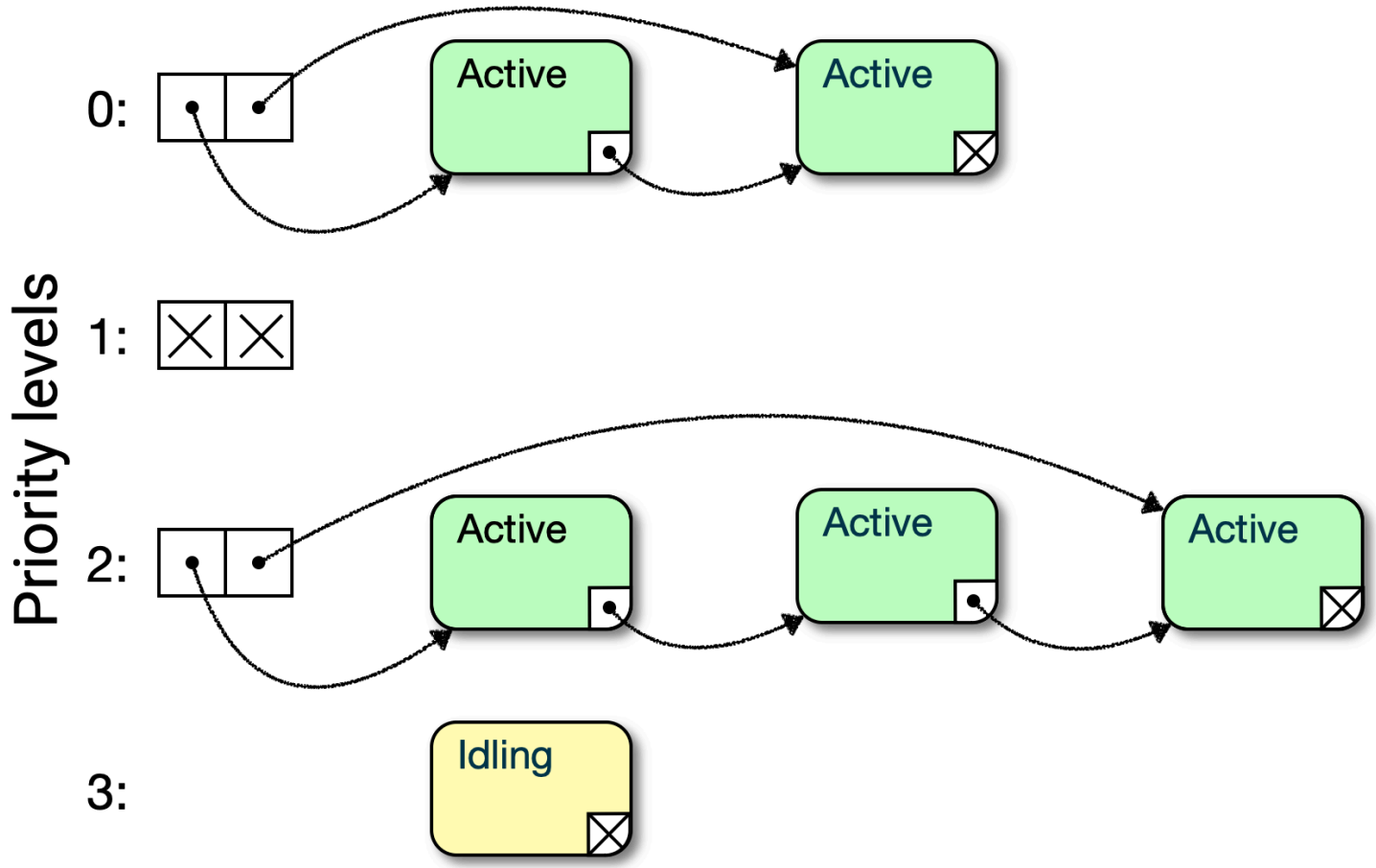
Draw memory layout (heap, stacks, ..., processes, main stack). ⟸

**? How does the OS decide which process to run next?**

Start with the easy case: context switch initiated by `yield()`.

# Ready Queues

In `ACTIVE` processes, the ready queue is implemented using the `next` field.



# Ready Queues

```
#define NPRIO 3       /* Number of non-idle priorities */
...
static struct _queue {
    proc head, tail;
} os_readyq[NPRIO];
```

Last line of `start(...)`, after setting up fake exception frame...

```
    make_ready(p);
```

Adds process to appropriate queue.

# Adding to the Queue: `make_ready()`

```
static inline void make_ready(proc p) {
    int prio = p->priority;
    if (prio == P_IDLE) return;

    p->state = ACTIVE;
    p->next = NULL;    // Because last in queue
    queue q = &os_readyq[prio];
    if (q->head == NULL)
        q->head = p;
    else
        q->tail->next = p;
    q->tail = p;                          }
```

# Full Story of Startup

Have code side-by-side.

- `startup.c`: `__reset()` is called.
- `microbian.c`: `__start()` is called.
- Calls `init()`, defined by "our" program.
- Sets up processes by calling `start()`.
- Creates handcrafted (rather than hardware) stack frame, so process can be started through normal return.
- Calls `make_ready()` to add process to appropriate queue.
  At initialisation, all processes are `ACTIVE` and in the ready queue.
- Make current function the idle task.
- `yield()`, so OS starts scheduling procedure.

# Selecting Next Process

We end up in `system_call(...)` again.

```
unsigned *system_call(unsigned *psp) {
    short *pc = (short *) psp[PC_SAVE]; /* Program counter */
    int op = pc[-1] & 0xff;  // Syscall number from SVC instr
    os_current->sp = psp;    // Save sp of the current process
    ...
    switch (op) {
    case SYS_YIELD:
        // Add current to back of Q. Choose next process.
        make_ready(os_current); choose_proc(); break;
        ... }
    return os_current->sp; }  // Return sp for next process
```

## Selecting Next Process

Find highest priority process to run, remove it from queue, `return`.

```
static inline void choose_proc(void) {
    for (int p = 0; p < NPRIO; p++) {
        queue q = &os_readyq[p];
        if (q->head != NULL) {
            os_current = q->head;
            q->head = os_current->next;
            return;
        }
    }
    os_current = idle_proc;
}
```

**? How does the OS decide which process to run next?**

But this time, if we send a message.

Remember: Send and receive are also calls to OS!

`microbian.c`:

```
void NOINLINE send(int dest, message *msg) {
    syscall(SYS_SEND);
}
```

`hardware.h`:

```
#define syscall(op)     asm volatile ("svc %0" : : "i"(op))
```

## System Call after `send()`

Just like `yield()`, `send()` simply causes a syscall interrupt.

```
unsigned *system_call(unsigned *psp) {
  short *pc = (short *) psp[PC_SAVE];
  int op = pc[-1] & 0xff;
  os_current->sp = psp;
  switch (op) {
  case SYS_SEND:
    mini_send(arg(0, int), arg(1, int), arg(2, message *));
    break; ...
  }
  return os_current->sp;
}
```

## Sending Messages

Remember the rules for communication by messages!

Two possible cases:

The destination process is `RECEIVING` and can receive message

- Transfer message to destination. Destination `READY`. Add to queue.

The destination process cannot receive…

- Multiple possible reasons for this (Q: which?)
- Must (somehow) keep sender in queue until process is ready to receive it.

# Sending Queues

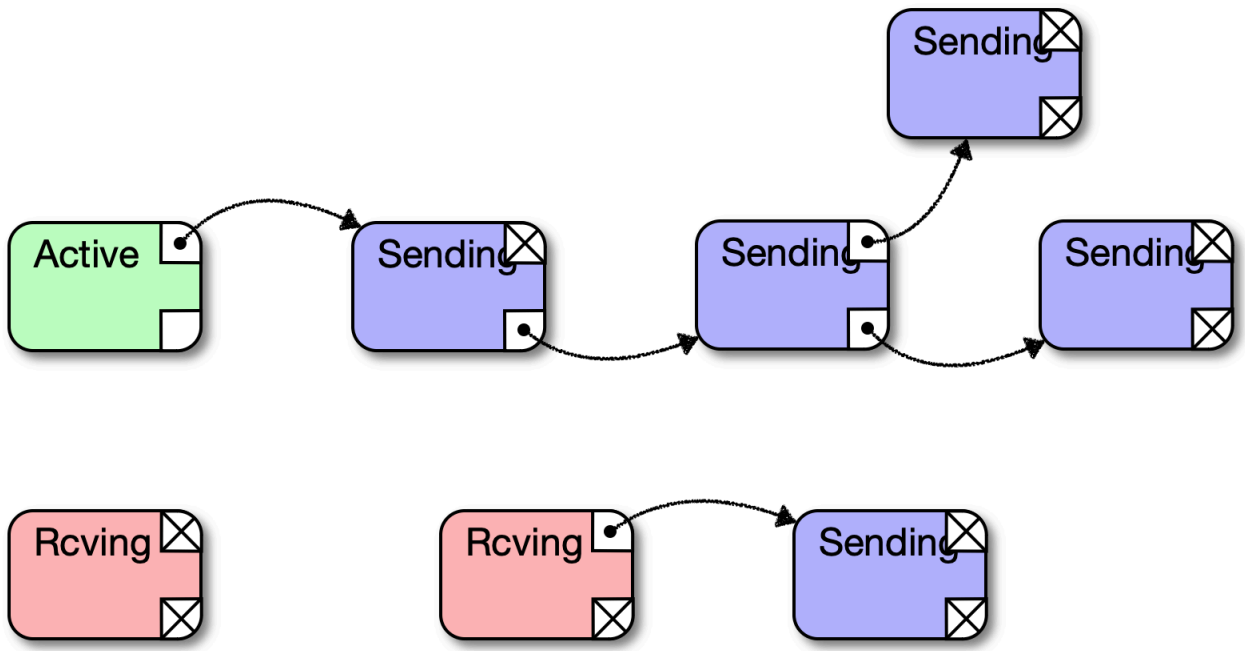
Processes can have multiple other processes waiting to send.

Next process trying to send is stored in `waiting`.

Rest of the send queue is chained through `next`.

Q: Does this conflict with the ready queue?

Each process has a queue of others waiting to send to it.



- Active process has 3 processes waiting to send to it.
- One process waiting to send also has a process waiting to send to *it*.
- One waiting to receive, but nothing to send to it yet.
- One waiting to receive, with one waiting to send to it, but not the right type of message.

# System call after send()

```
static void mini_send(int dest, int type, message *msg) {
  int src = os_current->pid;
  proc pdest = os_ptable[dest];
  if (accept(pdest, type)) { // Receiver is waiting
    deliver(pdest->msgbuf, src, msg);
    make_ready(pdest); make_ready(os_current);  // *DISCUSS*
  } else {
    os_current->state = SENDING;
    queue_send(pdest);  // Sender waits. Join receiver's queue
  }
  choose_proc();
}
```

**? How does the OS decide which process to run next?**

But this time, if we **receive** a message.

Three possible cases:

Can receive a message from an interrupt ⇒ discuss later.

Suitable message is waiting to be delivered.

- Deliver message of suitable type, from process.
- Both processes can now continue.

No suitable message can be delivered.

-

# Receiving Messages

```
static void mini_receive(int type, message *msg) {
  if (/* interrupt is due */) { // later...
  } else {
    proc psrc = find_sender(os_current, type);// search send Q
    if (psrc != NULL) { // Is a sender waiting?
      deliver(msg, psrc->pid, psrc->msgbuf);
      make_ready(os_current); make_ready(psrc);
    } else { // No luck: we must wait
      set_state(os_current, RECEIVING, type, msg);
  } }
  choose_proc();
}
```

**? How are messages transferred between processes?**

And in particular, how is this done in interrupts?

# Interrupt Handler

```
void default_handler(void) {
    int irq = active_irq(), task;
    if (irq < 0 || (task = os_handler[irq]) == 0)
        panic("Unexpected interrupt %d", irq);
    disable_irq(irq);
    interrupt(task);
}
```

Discuss code.

## How are messages delivered

```
static inline void deliver(proc pdest, proc psrc)
{
    if (pdest->msgbuf) {
        *(pdest->msgbuf) = *(psrc->msgbuf);
        pdest->msgbuf->sender = psrc->pid;
    }
    make_ready(pdest);
}
```

## Summary

- Process table
- Process states
- Ready queues
- Send queues
- Implementation of delivery schedule of messages

Notes contain another example of a device driver, and a discussion of `SENDREC`. Do take a look at this.

# End of Term!

Next term, start from the complete bottom of the stack:

**? How to we build a computer?**

... starting from a transistor